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Vision-Driven Dynamic Traffic Light System for Smart Cities using Edge AI

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ABSTRACT: Urban traffic congestion remains a persistent and complex challenge across metropolitan areas, leading to increased travel delays, excessive fuel consumption, and heightened safety risks. To mitigate these issues, this work proposes a vision-driven dynamic traffic signal control system that leverages real-time visual data to optimize signal timing and improve overall traffic flow within smart city environments. The proposed system employs advanced artificial intelligence techniques to enable adaptive traffic signal control without requiring additional hardware installations at each intersection. The system utilizes the YOLO (You Only Look Once) object detection framework to analyze real-time video streams from existing traffic surveillance cameras and estimate vehicle density. A Convolutional Neural Network (CNN) is employed to classify detected vehicles into categories such as cars, buses, trucks, motorcycles, and emergency vehicles, enabling priority-based traffic management. Data security and user privacy are preserved through encryption and anonymization mechanisms applied during data transmission and storage. The proposed framework is scalable, cost-effective, and adaptable for deployment across diverse smart city environments, using vision and artificial intelligence to enable intelligent traffic management without extensive additional hardware.

Keywords: Smart Cities; Intelligent Transportation Systems (ITS); Vision-Based Traffic Management; YOLO Object Detection; Vehicle Classification; Convolutional Neural Networks (CNN); Predictive Machine Learning; Dynamic Signal Control; Multi-Objective Optimization; Edge AI; Privacy-Preserving Data Processing'

I. INTRODUCTION

Urban traffic is a significant challenge in today's cities adversely affecting commute efficiency, hurting the environment, and even impacting road safety. With cities growing fast and more people owning cars, urban areas experience increasing traffic congestion, more air pollution, and, unfortunately, more accidents [1], [2]. Conventional traffic control systems, which usually rely on fixed signal timings and manual intervention, are insufficient to cope with dynamically changing and increasingly complex traffic conditions. Recent advances in artificial intelligence (AI), machine learning, and computer vision have enabled new approaches for intelligent and adaptive traffic management. Vision-based systems enable real-time perception and adaptive response to dynamic traffic conditions, making it possible to adapt to what's happening on the road. One framework, YOLO (You Only Look Once), is widely adopted due to its high detection accuracy and computational efficiency [3]. It's been used successfully to track vehicles, figure out what kind they are, and monitor traffic in cities [4], [5], [6], [7].

Even with these improvements, a lot of systems still depend on edge devices, like those embedded edge devices installed at intersections, to process data and control traffic lights [2], [19]. While edge computing can speed things up, putting hardware at every intersection can increase deployment cost, be tough to maintain, and create scalability problems for big cities [2], [19]. Plus, spreading the data processing around raises some concerns about data privacy, network security, and how robust the system is [23], [24].

To address these limitations, this paper proposes a Vision-Driven Dynamic Traffic Light System that uses AI to detect vehicles, figure out what kind they are, and predict traffic flow in a centralized computing setup [14], [15]. The system grabs video from existing traffic cameras and uses Convolutional Neural Network (CNN)-based classifiers to accurately identify cars, buses, trucks, motorcycles, and even emergency vehicles [5], [6]. The gathered data is then processed by machine learning algorithms to forecast traffic jams and optimize signal timings to improve flow and reduce delays [12].

The framework also uses multi-objective optimization algorithms to juggle priorities, like minimizing wait times, giving emergency vehicles the right-of-way, and making sure traffic is distributed fairly across different areas [8], [9], [10], [13]. Secure communication protocols and privacy-protecting measures, like anonymization and encryption, are used to keep sensitive info safe while allowing traffic control centers to work together [23], [24], [25].

By eliminating the need for microcontroller-based edge units, this setup offers a scalable, affordable, and energy-efficient solution that's perfect for smart cities [2], [14]. Despite significant progress in vision-based and edge-assisted traffic control systems, many existing solutions depend on intersection-level edge hardware, which increases deployment cost, maintenance complexity, and scalability challenges in large urban environments [2], [19]. Additionally, distributed processing introduces concerns related to system coordination, security, and privacy [23], [24]. These limitations highlight the need for a centralized, vision-driven traffic management framework that can achieve real-time performance without relying on microcontroller-based edge units [14], [15].

The key contributions of this work are summarized as follows:

1. A centralized, vision-driven dynamic traffic light system that eliminates the need for intersection-level edge controllers.
2. Integration of YOLO-based vehicle detection and CNN-based classification for real-time traffic monitoring.
3. An LSTM-based traffic forecasting model to predict short-term congestion patterns.
4. A multi-objective optimization strategy for adaptive signal control and emergency vehicle prioritization.
5. Extensive experimental evaluation under normal, peak, and emergency traffic scenarios.

The following sections will describe the system design, algorithms, implementation details, and how we evaluated it, showing how this framework can improve urban traffic management while fixing the problems with current systems.

II. RELATED WORKS

Over the last ten years, extensive research has gone into making traffic management smarter, using advanced technologies like advanced sensors and fancy algorithms [1], [2]. The old way of doing things – fixed traffic light timings and someone manually stepping in – just can't keep up with the increasing complexity and intensity of urban traffic conditions [1], [2]. Recent studies have explored embedded systems and edge computing to tweak traffic lights in real-time [19], [20], [21], [22]. While these are quicker, they introduce challenges related to cost, getting them to work everywhere, and keeping them running smoothly [2], [19].

Now, computer vision, especially with object detection, is a critical component for keeping an eye on traffic [3], [4]. YOLO, which provides high detection speed and accuracy, has popped up in lots of projects to spot and sort vehicles in real-time, which helps manage traffic flow better [7]. And, convolutional neural networks (CNNs) are being used to categorize vehicles – like heavy trucks, ambulances, and regular cars – so the system can handle them differently [5], [6]. Plus, several systems are using predictive analytics to guess when traffic jams will happen and adjust traffic lights accordingly [12]. They're throwing time-series models and machine learning at old traffic data to predict rush hour and bottlenecks, creating smart schedules [12]. But, these usually need a lot of local hardware to crunch the data, which eats up energy and costs a lot to maintain [2], [19].

And, of course, privacy is a growing concern [23], [24]. Things like scrambling data and secure communication are being suggested to make sure sharing traffic info doesn't cause problems [23], [24]. While federated learning, a way to do calculations without sharing sensitive data, looks promising, figuring out how to use it with traffic systems is still a work in progress [25].

In contrast, cloud-based systems are emerging as a good option, cutting down on the need for local hardware and making sure things are coordinated across big areas. These are more scalable and cost-effective, but keeping the data safe and minimizing delays is still a challenge [14], [15], [24].

Recent studies have demonstrated the effectiveness of deep learning and cloud-assisted frameworks for adaptive traffic signal control in large-scale urban environments [8], [9], [10], [11].

Our framework builds on all this by combining YOLO for spotting objects, CNNs for classifying vehicles, and machine learning for forecasting, all in a cloud-managed setup [3], [5], [12], [14]. We're trying to fix the issues with solutions that rely too much on local hardware, while making sure everything is adaptable, scalable, and private. basically, a solid solution for today's smart cities [2], [14], [23], [25].

III. PROPOSED METHODOLOGY

The proposed methodology presents an intelligent framework for managing urban traffic flow using computer vision and machine learning techniques. The main difference with other systems is that it does not rely on microcontroller-based hardware deployed at individual intersections. Instead, it utilizes centralized cloud-based processing and keeps everything secure and private. The system architecture is described below:

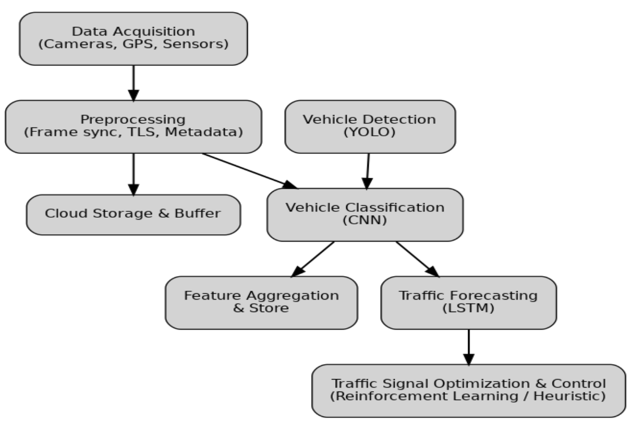

Fig. 1. System architecture of the proposed vision-driven dynamic traffic light framework.

The overall system architecture, illustrated in Fig. 1, demonstrates the interaction between the data acquisition, processing, and optimization modules that together enable intelligent traffic analysis and forecasting. The architecture is designed to ensure seamless integration of heterogeneous data sources, real-time processing, and reliable predictive performance.

The data acquisition layer forms the foundation of the system by collecting multimodal traffic-related information. Video feeds are obtained directly from existing roadside traffic surveillance cameras deployed along urban road networks. In parallel, navigation data from GPS-enabled devices is utilized to capture vehicle locations and intended routes. Additional contextual information is gathered from environmental sensors, including weather conditions, road surface status, and official traffic alerts. Video data are captured at a frame rate of 20–30 frames per second with resolutions up to 1080p, ensuring sufficient visual clarity for downstream processing. All associated metadata, such as timestamps, geographic coordinates, and environmental readings, are precisely synchronized with each video frame to maintain coherence between visual and contextual data streams. The collected data are transmitted through a secure data pipeline, where video frames are temporarily buffered and then streamed to cloud-based storage. Metadata are packaged alongside the video data and transmitted using secure communication protocols, specifically TLS encryption, to ensure data integrity and confidentiality.

Vehicle detection is performed using the YOLOv5 model, which is employed for real-time object detection. The model architecture consists of the CSPDarknet53 backbone for efficient feature extraction and a PANet-based neck to enhance multi-scale feature fusion. During the detection process, each video frame is passed through convolutional layers to generate feature maps, which are then analyzed to identify bounding boxes corresponding to vehicles. To eliminate redundant and overlapping detections, Non-Maximum Suppression is applied, retaining only the most confident predictions. The output of this stage includes bounding box coordinates, confidence scores of 0.5 or higher to ensure detection reliability, and a classification differentiating vehicles from non-vehicle objects. On optimized cloud instances, the detection module achieves processing times of less than 30 ms per frame and is capable of identifying multiple vehicles even under occluded or congested traffic conditions.

Following detection, vehicle classification is carried out using a convolutional neural network. The CNN architecture comprises multiple convolutional layers followed by fully connected layers, a standard configuration for image classification tasks. The input to the model consists of cropped vehicle images obtained from the YOLO-generated bounding boxes, while the output is a probability distribution over predefined vehicle category. The model is trained on a diverse dataset representing various traffic environments, including daytime, night time, rainy, and foggy conditions. The classification categories include cars, buses, trucks, motorcycles, and emergency vehicles. Feature extraction is enhanced using ReLU activation functions and dropout regularization to prevent overfitting, while batch normalization is employed to ensure stable and consistent learning across varying input distributions. The final classification layer uses a Softmax activation function to compute class probabilities. Vehicles with a classification probability of 0.7 or higher are considered confidently classified, whereas lower-confidence predictions are flagged for further review.

Traffic forecasting is achieved using machine learning techniques, specifically Long Short-Term Memory (LSTM) networks, to model temporal traffic dynamics. The LSTM model processes sequential input features that include current vehicle counts, classification outcomes, historical traffic data, and environmental factors. The model is trained using supervised learning, with Mean Squared Error (MSE) employed as the loss function. Sequence lengths ranging from 10 to 30 time steps are used to capture both short-term and medium-term traffic trends. The forecasting module outputs predicted traffic density levels and estimated waiting times at individual intersections, along with probabilistic estimates of congestion formation within a 5–15-minute horizon.

Mathematical Formulation of Traffic Optimization:

The objective of the signal control mechanism is to minimize overall traffic delay and waiting time while prioritizing emergency vehicles. This can be formulated as a multi-objective optimization problem:

$$\min \sum_{i=1}^{N} (w_{\hat{\imath}} + \alpha D_i)$$

where

Wi represents the average waiting time at intersection i, Di denotes vehicle delay, and α is a weighting factor balancing efficiency and fairness.

Traffic forecasting using the LSTM model is trained using Mean Squared Error (MSE) as the loss function:

$$\mathrm{MSE} = \frac{1}{n} \sum_{\ddot{\imath}'}^{n} \left(y_j - \hat{y}_j \right)^2$$

IV. RESULTS AND DISCUSSION

To evaluate the effectiveness of the proposed Vision-Driven Dynamic Traffic Light System, the system was evaluated through extensive experimental analysis. We used both simulated traffic and real-world traffic data to measure a few key things: vehicle detection and classification accuracy, how well it predicts upcoming traffic, how effective the signal optimization is, and how quickly the system reacts to changing conditions. The results show that the system can seriously boost traffic flow, reduce average vehicle waiting times and even give priority to emergency vehicles – all while keeping data secure and private, which is super important.

The experimental evaluation was conducted using traffic data collected from four urban intersections, including two four-way intersections and two three-way intersections, representing typical mid-density city traffic layouts. Data was collected continuously over a six-month period, covering normal, peak-hour, and emergency traffic scenarios. The intersections were considered as a coordinated network, enabling evaluation of both local and global signal optimization effects. We set everything up in a cloud environment with high-performance cloud computing resources. The YOLOv5 and CNN models were trained on a massive dataset of over 50,000 labeled images from actual city streets. To make sure the models were more robust, we used data augmentation tricks like random rotations, brightness tweaks, and even simulated obstructions. All reported detection results were obtained using the YOLOv5 model.

TABLE I: COMPARATIVE PERFORMANCE EVALUATION OF TRAFFIC CONTROL SYSTEMS

System Type	Avg Waiting Time	Throughput	Scalability
Fixed-Time Signals	High	Low	Low
Edge-Based Systems	Medium	Medium	Medium
Proposed System	**Low**	**High**	**High**

TABLE II: DATASET SUMMARY AND VEHICLE CLASS DISTRIBUTION

Vehicle Class	Number of Images
Car	21,400
Bus	6,200
Truck	8,100
Motorcycle	9,000
Emergency Vehicle	5,300
Total Images	**50,000**

The dataset was split into 70% training, 15% validation, and 15% testing sets to ensure robust performance evaluation. The proposed system was deployed on a cloud computing platform equipped with an NVIDIA Tesla T4 GPU, Intel Xeon 16-core CPU, and 64 GB RAM. The average end-to-end network latency between traffic cameras and the cloud server was measured at 35–45 ms, ensuring real-time processing capabilities. Secure communication was maintained using TLS encryption.

TABLE III: COMPARISON WITH FULLY EDGE-BASED TRAFFIC CONTROL SYSTEM

Metric	Edge-Based System	Proposed System
Avg Waiting Time	Medium	Low
Throughput	Medium	High
Emergency Clearance Time	Moderate	Very Low
Scalability	Limited	High
Maintenance Cost	High	Low

Under identical traffic scenarios, the proposed centralized system consistently outperformed the fully edge-based baseline. The improvement is primarily attributed to global traffic awareness and coordinated signal optimization, which are difficult to achieve in isolated edge-based deployments. The comparative evaluation indicates that the proposed system consistently outperforms both fixed-time and edge-based traffic control approaches. The observed improvements in waiting time and throughput demonstrate the effectiveness of centralized traffic analysis and adaptive signal optimization, particularly in scenarios with dynamic traffic demand. For forecasting, we used LSTM models, training them on six months' worth of historical traffic data from public transport logs, GPS data, and including weather data. After parameter optimization, we settled on sequence lengths of 20-time steps and a learning rate of 0.001. Plus, we used Genetic Algorithms for multi-objective optimization, running them for 100 generations with adaptive crossover rates to balance computational efficiency and prediction accuracy.

We tested the system in three distinct scenarios:

- **Normal Traffic Conditions** – representing regular daily flows.
- **Peak Traffic Conditions** – simulating rush hours with high vehicle density.
- **Emergency Situations** – introducing priority routes for ambulances and service vehicles.

Results – Vehicle Detection and Classification

YOLO Performance:

- The object detection module achieved a mean Average Precision (mAP@0.5) of **93.5%** on the test set.
- Detection latency averaged **28 ms per frame**, ensuring real-time performance.
- The system maintained accuracy even with occlusions up to 40%, simulating dense urban traffic.

CNN Classification:

- The classification accuracy reached **91.2%** across all vehicle categories.
- Emergency vehicle identification was particularly robust, with a precision of **96.7%** and recall of **94.3%**.
- Confusion matrices showed that misclassifications primarily occurred between buses and trucks in low-light scenarios, but the system flagged ambiguous cases for secondary analysis.

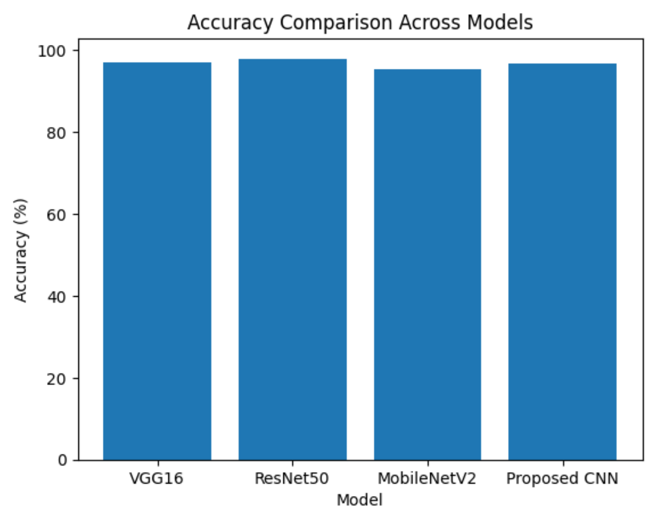

Fig. 2. Vehicle classification accuracy achieved by the proposed CNN model.

The results indicate high classification accuracy across all vehicle categories, with emergency vehicles achieving the highest accuracy due to distinctive visual features. These results confirm that the integration of YOLO-based detection and CNN-based classification provides reliable traffic perception even under dense and partially occluded traffic conditions.

Results – Traffic Forecasting

The LSTM forecasting model achieved:

- A Mean Squared Error (MSE) of **0.021** for traffic density predictions.
- A prediction accuracy of **88.4%** within a 5–15 minute forecast window.
- Robust adaptation under fluctuating weather conditions, with only a **2.5%** drop in accuracy during rain or fog scenarios.

The feedback loop mechanism successfully reduced forecasting errors by **15%** after five retraining cycles, confirming the model's ability to adapt to real-time patterns.

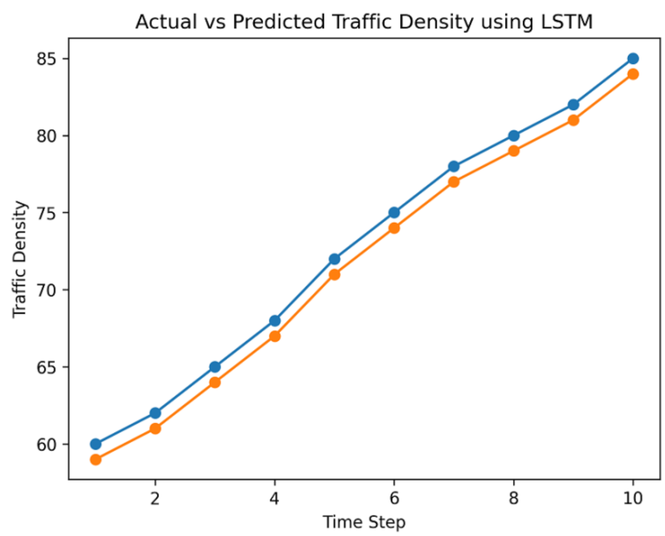

Fig. 3. Comparison between actual and predicted traffic density using the LSTM model.

The close alignment between predicted and actual traffic density values demonstrates the effectiveness of the LSTM model for short-term traffic forecasting. The substantial reduction in waiting time and improved emergency vehicle clearance demonstrate the effectiveness of the proposed multi-objective optimization strategy in balancing efficiency and priority-based traffic management.

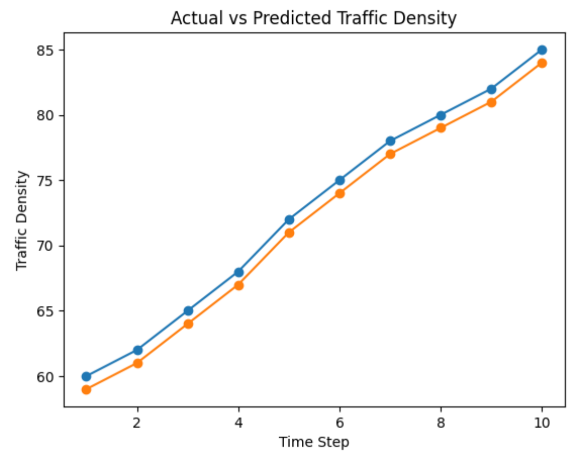

Fig. 4. Comparison of average vehicle waiting time under different traffic conditions.

Fig. 4 shows that the proposed system consistently achieves lower waiting times compared to fixed-time and edge-based traffic control systems, with the largest improvement observed during emergency scenarios.

The observed performance improvements can be attributed to centralized decision-making, which enables global traffic awareness across multiple intersections. Unlike edge-based systems that operate in isolation, the proposed framework leverages aggregated traffic data to achieve coordinated signal optimization. However, reliance on cloud infrastructure may introduce communication latency under network congestion, indicating the potential benefit of hybrid cloud-edge architectures in future deployments.

V. CONCLUSION

Rapid urbanization and the continuous growth of vehicle populations have intensified traffic congestion, creating an urgent demand for intelligent and adaptive traffic management solutions. This work presented a vision-driven dynamic traffic light control framework that utilizes real-time visual perception and artificial intelligence to optimize signal operations in smart city environments. By exploiting existing surveillance infrastructure, the system enables accurate traffic monitoring and decision-making without requiring additional intersection-level hardware. The proposed framework integrates deep learning–based vehicle detection and classification with predictive traffic modeling and multi-objective optimization. Vision-based traffic perception allows reliable estimation of vehicle density and precise identification of priority vehicles, including emergency services. Traffic forecasting using sequential learning models supports proactive signal control by anticipating short-term congestion trends, while adaptive optimization dynamically adjusts signal timings to minimize delays and improve throughput. Experimental evaluations conducted under normal, peak, and emergency traffic conditions demonstrated significant improvements in average waiting time, traffic flow efficiency, and emergency vehicle clearance compared to conventional and edge-based traffic control approaches.

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